

Q No. 1
Question

If the number $27215*2$ is completely divisible by 11, then the value of $*$ is :

- A) 3
- B) 2
- C) 1
- D) 5

Correct Answer

3

Solution

For a number to be divisible by 11,

$$2 + 2 + 5 + 2 = 11 \quad 7 + 1 + * = 8 + *$$

$11 - (8 + *)$ has to be a multiple of 11

$$\therefore * = 3.$$

Q No. 2
Question

If $x + \frac{1}{x} = 3$ then value of $\sqrt{x} + \frac{1}{\sqrt{x}}$ is

- A) $\sqrt{5}$
- B) $\sqrt{2}$
- C) $\sqrt{3}$
- D) $\sqrt{7}$

Correct Answer

$\sqrt{5}$

Solution

$$\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2} = \sqrt{x + \frac{1}{x} + 2} = \sqrt{5}$$

Q No. 3
Question

If $a + b + c = 0$, then $\frac{1}{3} \left(\frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} \right)$ is

- A) 0
- B) $\frac{1}{2}$
- C) 1
- D) $\frac{3}{2}$

Correct Answer

1

Solution

$$a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$$

$$\text{Now, } \frac{1}{3} \left(\frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} \right) = \frac{1}{3} \left(\frac{a^3 + b^3 + c^3}{abc} \right) = \frac{3}{3} = 1$$

Q No. 4
Question

The rational form of $2.74\overline{35}$ is

- A) $\frac{27161}{999}$
- B) $\frac{27}{99}$
- C) $\frac{27161}{9900}$
- D) $\frac{27161}{9000}$

Correct Answer

$$\frac{27161}{9900}$$

Solution

$$x = 2.74\overline{35} = 2.74353535$$

$$100x = 274.\overline{35} \quad \dots \text{ (i)}$$

$$10000x = 27435.\overline{35} \quad \dots \text{ (ii)}$$

$$\text{(ii)} - \text{(i)}$$

Q No. 5
Question

If $a : b = 3 : 2$, $b : c = 4 : 1$, $c : d = 5 : 2$, then $a : b : c : d$ is

- A) $2 : 5 : 20 : 30$
- B) $5 : 2 : 20 : 30$
- C) $1 : 2 : 3 : 4$
- D) $30 : 20 : 5 : 2$

Correct Answer

$30 : 20 : 5 : 2$

Solution

$$\begin{array}{cccc} a : b & : & c & : d \\ 3 : 2 & : & 2 & : 2 \\ 4 : 4 & : & 1 & : 1 \\ 5 : 5 & : & 5 & : 2 \\ \hline 60 : 40 & : & 10 & : 4 \\ 30 : 20 & : & 5 & : 2 \end{array}$$

Q No. 6
Question

If $\frac{x}{y} = \frac{1}{2}$, then find $\frac{4x^2 - 7y^2}{8x^2 + 3y^2}$

- A) $-6 : 5$
- B) $6 : 5$
- C) $5 : 6$
- D) $-5 : 6$

Correct Answer

$-6 : 5$

Solution

$$\frac{x}{y} = \frac{1}{2} \Rightarrow \frac{x^2}{y^2} = \frac{1}{4}$$

$$\text{Now, } \frac{4x^2 - 7y^2}{8x^2 + 3y^2} \Rightarrow \frac{4\left(\frac{x^2}{y^2}\right) - 7}{8\left(\frac{x^2}{y^2}\right) + 3} = \frac{4 \cdot \frac{1}{4} - 7}{8 \cdot \frac{1}{4} + 3} = \frac{-6}{5}$$

Q No. 7
Question

If the expression $2x^3 + 3px^2 - 4x + p$ has a remainder of 5 when divided by $x + 2$, then value of remainder when divided by $x - 1$ is ?

- A) 0
- B) 1
- C) 2
- D) 3

Correct Answer

2

Solution

Hint : Let $f(x) = 2x^3 + 3px^2 - 4x + p$

$$f(-2) = 5 \Rightarrow p = 1$$

Q No. 8
Question

If $x^{101} + 1$ is divided by $(x - 1)$ then remainder is?

- A) 3
- B) 2
- C) 1
- D) 0

Correct Answer

2

Solution

$$\text{Remainder} = 1^{101} + 1 = 2$$